1 Kinematics

$$\bar{u}_\alpha = u_\alpha + x_3\phi_\alpha, \qquad \bar{u}_3 = w,$$

$$\epsilon_{\alpha\beta} = \varepsilon_{\alpha\beta} + x_3\kappa_{\alpha\beta},$$

$$\varepsilon_{\alpha\beta} = \frac{1}{2}(u_{\alpha,\beta} + u_{\beta,\alpha}), \qquad \kappa_{\alpha\beta} = -\frac{1}{2}(\phi_{\alpha,\beta} + \phi_{\beta,\alpha}),$$

$$\gamma_\alpha = w_{,\alpha} - \phi_\alpha.$$

2 Constitutive relations

$$\sigma_{\alpha\beta} = D_{\alpha\beta\gamma\eta}\,\epsilon_{\gamma\eta}, \qquad \sigma_{3\alpha} = kG\,\gamma_\alpha,$$

$$D_{\alpha\beta\gamma\eta} = \frac{E}{1-\nu^2}\Big(\nu\delta_{\alpha\beta}\delta_{\gamma\eta} + \frac{1-\nu}{2}(\delta_{\alpha\gamma}\delta_{\beta\eta} + \delta_{\alpha\eta}\delta_{\beta\gamma})\Big),$$

$$G = \frac{E}{2(1+\nu)}.$$

3 Stress resultants

$$\bar{N}_{\alpha\beta} = \int_{-h/2}^{h/2} \sigma_{\alpha\beta} dx_3 = hD_{\alpha\beta\gamma\eta}\,\varepsilon_{\gamma\eta},$$

$$M_{\alpha\beta} = \int_{-h/2}^{h/2} x_3\sigma_{\alpha\beta} dx_3 = \frac{h^3}{12}D_{\alpha\beta\gamma\eta}\,\kappa_{\gamma\eta},$$

$$Q_\alpha = k\int_{-h/2}^{h/2} \sigma_{3\alpha} dx_3 = kGh\,\gamma_\alpha.$$

4 Equilibrium equations and boundary conditions (strong form

$$\delta\Pi = \int_\Omega \Big(\bar{N}_{\alpha\beta}\,\delta\varepsilon_{\alpha\beta} + M_{\alpha\beta}\,\delta\kappa_{\alpha\beta} + Q_\alpha\,(\delta\phi_\alpha + \delta w_{,\alpha}) - N_{\alpha\beta}\,w_{,\alpha}\delta w_{,\beta}\Big) d\Omega = 0.$$

Integration by parts yields:

$$\begin{aligned} &\bar{N}_{\alpha\beta,\beta} = 0 \quad \text{in } \Omega, \\ &M_{\alpha\beta,\beta} - Q_\alpha = 0 \quad \text{in } \Omega, \\ &Q_{\alpha,\alpha} - (N_{\alpha\beta}w_{,\beta})_{,\alpha} = 0 \quad \text{in } \Omega. \end{aligned}$$

Natural boundary conditions:

$$\bar{N}_{\alpha\beta}n_\beta = \bar{N}_\alpha \;\; \text{on } \Gamma_N, \qquad M_{\alpha\beta}n_\beta = \bar{M}_\alpha \;\; \text{on } \Gamma_M, \qquad (Q_\alpha - N_{\alpha\beta}w_{,\beta})n_\alpha = \bar{Q} \;\; \text{on } \Gamma_Q.$$

5 Variational formulation (weak form)

The weak form is obtained without integration by parts:

$$\int_{\Omega} \Big(\bar{N}_{\alpha\beta}\, \delta\varepsilon_{\alpha\beta} + M_{\alpha\beta}\, \delta\kappa_{\alpha\beta} + Q_{\alpha}\, (\delta\phi_{\alpha} + \delta w_{,\alpha}) - N_{\alpha\beta}\, w_{,\alpha}\delta w_{,\beta} \Big) d\Omega = 0.$$

Explicit variation with respect to (u_{\alpha}, w, \phi_{\alpha}):

$$\begin{aligned}
\delta u_{\alpha}: &\quad \int_{\Omega} \bar{N}_{\alpha\beta}\, \delta u_{\alpha,\beta}\, d\Omega = 0,\\
\delta \phi_{\alpha}: &\quad \int_{\Omega} \big(M_{\alpha\beta}\, \delta\phi_{\alpha,\beta} + Q_{\alpha}\, \delta\phi_{\alpha} \big) d\Omega = 0,\\
\delta w: &\quad \int_{\Omega} \big(Q_{\alpha}\, \delta w_{,\alpha} - N_{\alpha\beta} w_{,\beta}\, \delta w_{,\alpha} \big) d\Omega = 0.
\end{aligned}$$

Bilinear forms:

$$a_m = \int_{\Omega} h D_{\alpha\beta\gamma\eta}\, \varepsilon_{\alpha\beta}\delta\varepsilon_{\gamma\eta} d\Omega, \quad a_b = \int_{\Omega} \frac{h^3}{12} D_{\alpha\beta\gamma\eta}\, \kappa_{\alpha\beta}\delta\kappa_{\gamma\eta} d\Omega,$$

$$a_s = \int_{\Omega} kGh\, \gamma_{\alpha}\delta\gamma_{\alpha} d\Omega, \quad a_g = \int_{\Omega} N_{\alpha\beta}\, w_{,\alpha}\delta w_{,\beta} d\Omega,$$

$$a_m + a_b + a_s - \lambda a_g = 0.$$

6 Finite element discretization

$$u_{\alpha}^h = \sum_I N_I u_{\alpha I}, \qquad w^h = \sum_I N_I w_I, \qquad \phi_{\alpha}^h = \sum_I N_I \phi_{\alpha I},$$

$$\varepsilon_{\alpha\beta I} = \frac{1}{2}(N_{I,\beta} u_{\alpha I} + N_{I,\alpha} u_{\beta I}), \qquad \kappa_{\alpha\beta I} = -\frac{1}{2}(N_{I,\beta}\phi_{\alpha I} + N_{I,\alpha}\phi_{\beta I}),$$

$$\gamma_{\alpha I} = N_{I,\alpha} w_I - N_I \phi_{\alpha I}, \qquad w_{,\alpha I} = N_{I,\alpha} w_I.$$

$$K_{IJ}^m = \int_{\Omega} h D_{\alpha\beta\gamma\eta}\, \varepsilon_{\alpha\beta I}\, \varepsilon_{\gamma\eta J}\, d\Omega,$$

$$K_{IJ}^b = \int_{\Omega} \frac{h^3}{12} D_{\alpha\beta\gamma\eta}\, \kappa_{\alpha\beta I}\, \kappa_{\gamma\eta J}\, d\Omega,$$

$$K_{IJ}^s = \int_{\Omega} kGh\, \gamma_{\alpha I}\, \gamma_{\alpha J}\, d\Omega,$$

$$K_{IJ}^g = \int_{\Omega} N_{\alpha\beta}\, N_{I,\alpha} N_{J,\beta}\, d\Omega.$$

$$K_{IJ} = K_{IJ}^m + K_{IJ}^b + K_{IJ}^s.$$

$$\sum_J K_{IJ}\, \mathbf{d}_J = \lambda \sum_J K_{IJ}^g\, \mathbf{d}_J, \qquad \mathbf{d}_J = [u_{1J}, u_{2J}, w_J, \phi_{1J}, \phi_{2J}]^T.$$

$$\boxed{(\mathbf{K} - \lambda \mathbf{K}_G)\, \mathbf{d} = \mathbf{0}.}$$